EXPERIMENT NO 02

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SLOT: A

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SUBJECT: DEEP LEARNING

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Covariance, Eigenvalues and Eigenvectors Analysis

SCENARIO

A healthcare dataset contains patients' height, weight, and blood pressure. Calculate the covariance matrix to identify relationships among the features and visualize the correlations.

AIM

To calculate the covariance matrix of patients' height, weight, and blood pressure using Python and NumPy, determine the eigenvalues and eigenvectors, and visualize the relationships among the features.

ALGORITHM

1. Import the required libraries.
2. Create a healthcare dataset containing height, weight, and blood pressure.
3. Calculate the covariance matrix.
4. Compute the eigenvalues and eigenvectors of the covariance matrix.
5. Visualize the feature relationships using a heatmap.
6. Display the results and interpret the relationships.

SOURCECODE

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Create healthcare dataset (sample data)
df = pd.DataFrame({
   'Height': [150, 160, 165, 170, 175, 180],
```

```
    'Weight': [50, 55, 60, 68, 72, 80],
    'Blood_Pressure': [110, 115, 120, 125, 130, 140]
})

# Display dataset and summary
print("Healthcare Dataset")
display(df)
print("Summary Statistics")
display(df.describe().round(2))

# Calculate covariance matrix
cov_matrix = df.cov()
print("Covariance Matrix")
display(cov_matrix.round(2))

# Calculate eigenvalues and eigenvectors
eigenvalues, eigenvectors = np.linalg.eig(cov_matrix)
print("Eigenvalues")
print(np.round(eigenvalues, 2))
print("\nEigenvectors")
print(np.round(eigenvectors, 3))

# Visualize covariance matrix using a heatmap
plt.figure(figsize=(7, 5))
plt.imshow(cov_matrix, cmap='coolwarm', aspect='auto')
plt.colorbar(label='Covariance')
plt.xticks(range(3), df.columns, rotation=20)
plt.yticks(range(3), df.columns)
plt.title('Covariance Matrix Heatmap')
```

```
for i in range(3):
    for j in range(3):
        plt.text(j, i, f'{cov_matrix.iloc[i, j]:.1f}',
                 ha='center', va='center')
plt.tight_layout()
plt.show()

# Visualize feature relationships
pd.plotting.scatter_matrix(
    df, figsize=(9, 7), diagonal='hist',
    color='blue', alpha=0.7
)
plt.suptitle('Relationships Among Healthcare Features')
plt.tight_layout()
plt.show()
```

OUTPUT

Healthcare Dataset

	Height	Weight	Blood_Pressure
0	150	50	110
1	160	55	115
2	165	60	120
3	170	68	125
4	175	72	130
5	180	80	140

Summary Statistics

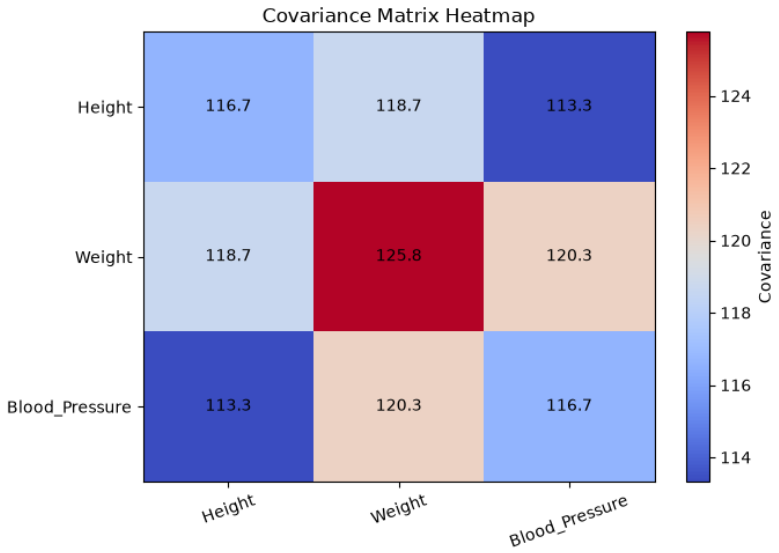
	Height	Weight	Blood_Pressure
count	6.00	6.00	6.00
mean	166.67	64.17	123.33
std	10.80	11.21	10.80
min	150.00	50.00	110.00
25%	161.25	56.25	116.25
50%	167.50	64.00	122.50
75%	173.75	71.00	128.75
max	180.00	80.00	140.00

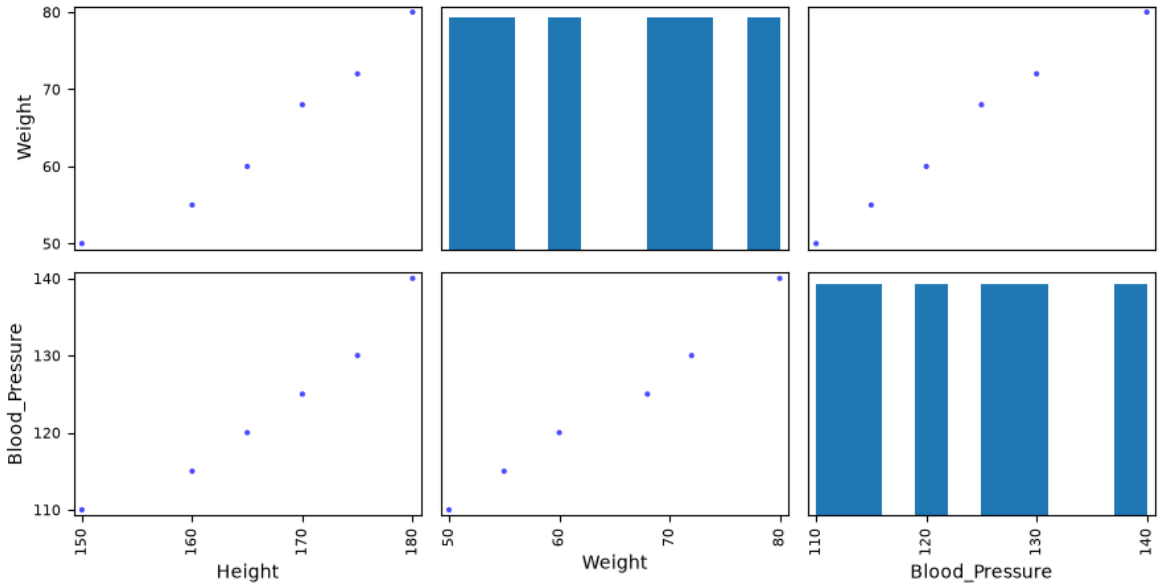
Covariance Matrix

	Height	Weight	Blood_Pressure
Height	116.67	118.67	113.33
Weight	118.67	125.77	120.33
Blood_Pressure	113.33	120.33	116.67

Eigenvalues
[354.74 3.67 0.69]

Eigenvectors
[[-0.568 -0.808 0.156]
[-0.594 0.271 -0.758]
[-0.57 0.523 0.634]]





RESULT

Thus, the covariance matrix, eigenvalues, and eigenvectors were successfully calculated using Python. The heatmap and scatter plots help visualize relationships among the healthcare features, while eigenvalues and eigenvectors identify the principal directions of variation in the dataset. The analysis demonstrates the application of linear algebra in healthcare data analysis.